

Conception

Pregnancy begins with conception, a complex process that involves the release of one or more eggs from the ovary, successful fertilization by a sperm in a fallopian tube, and implantation in the lining of the uterus.

The release of an egg

Each woman is born with her full quota of follicles that contain immature eggs, some of which will mature and be released in her lifetime. Every month, follicle-stimulating hormone (FSH), released from the pituitary gland, encourages a number of the follicles to ripen. These follicles in turn produce the hormone estrogen, rising levels of which encourage the uterus to thicken to prepare for the implantation of a fertilized egg. As the eggs mature, the level of estrogen rises and the pituitary gland receives a message to produce luteinizing hormone (LH). Every month, this surge in LH triggers one follicle (and sometimes more than one) to release a mature egg—the moment of ovulation.

Once the egg leaves the ovary, it enters the fallopian tube, which lies close by, and starts to travel through the tube to the uterus. The fallopian tube is just 4 in (10 cm) long and its lining has many tiny fronds that literally brush the egg in the direction of the uterus. Even so, the journey takes five days or more. In the course of this voyage, the fertilization of the egg takes place.